

Exp 9 Given Data

Generator cost equations:

$$F_i = a_i + b_i P_i + c_i P_i^2$$

Generator	a	b	c	Pmin	Pmax
G1	500	5.3	0.004	100	600
G2	400	5.5	0.006	100	400
G3	200	5.8	0.009	50	200

Total Demand:

$$P_D = 850MW$$

Step 1: Economic Dispatch Equation

Incremental cost equation:

$$\lambda = \frac{dF}{dP} = b + 2cP$$

Therefore,

$$P = \frac{\lambda - b}{2c}$$

For each generator:

Generator 1

$$P_1 = \frac{\lambda - 5.3}{0.008}$$

Generator 2

$$P_2 = \frac{\lambda - 5.5}{0.012}$$

Generator 3

$$P_3 = \frac{\lambda - 5.8}{0.018}$$

Step 2: Apply Power Balance Equation

$$P_1 + P_2 + P_3 = 850$$

Substitute equations:

$$\frac{\lambda - 5.3}{0.008} + \frac{\lambda - 5.5}{0.012} + \frac{\lambda - 5.8}{0.018} = 850$$

Multiply throughout by **72**

$$9000(\lambda - 5.3) + 6000(\lambda - 5.5) + 4000(\lambda - 5.8) = 61200$$

Expand:

$$\begin{aligned} &9000\lambda - 47700 \\ &+6000\lambda - 33000 \\ &+4000\lambda - 23200 \\ &= 61200 \end{aligned}$$

Combine:

$$19000\lambda - 103900 = 61200$$

Add:

$$\begin{aligned} 19000\lambda &= 165100 \\ \lambda &= 8.689 \end{aligned}$$

After MATLAB iterative correction and limit checking:

$$\boxed{\lambda \approx 8.69}$$

Step 3: Calculate Generator Outputs

Generator 1

$$P_1 = \frac{8.69 - 5.3}{0.008}$$
$$P_1 = 423.68MW$$

Generator 2

$$P_2 = \frac{8.69 - 5.5}{0.012}$$
$$P_2 = 265.79MW$$

Generator 3

$$P_3 = \frac{8.69 - 5.8}{0.018}$$
$$P_3 = 160.53MW$$

Step 4: Verify Total Generation

$$P_T = 423.68 + 265.79 + 160.53$$
$$P_T = 850MW$$

Hence,

$$\boxed{P_G = P_D}$$

Condition satisfied.

Step 5: Calculate Total Operating Cost

Generator 1 Cost

$$F_1 = 500 + 5.3(423.68) + 0.004(423.68)^2$$
$$F_1 = 500 + 2245.50 + 718.82$$
$$F_1 = 3464.32$$

Generator 2 Cost

$$F_2 = 400 + 5.5(265.79) + 0.006(265.79)^2$$

$$F_2 = 400 + 1461.85 + 423.88$$

$$F_2 = 2285.73$$

Generator 3 Cost

$$F_3 = 200 + 5.8(160.53) + 0.009(160.53)^2$$

$$F_3 = 200 + 931.07 + 231.12$$

$$F_3 = 1362.19$$

Step 6: Total Cost

$$F_T = 3464.32 + 2285.73 + 1362.19$$

$$\boxed{F_T = 7112.24}$$